Fiscal Year:	FY 2004	**Task Last Updated:**	FY 03/17/2006
PI Name:	Anderson, Mark S		
Project Title:	Biomolecular Imaging with Atomic Force Microscope-Mediated Raman Spectroscopy		
Division Name:	Physical Sciences		
Program/Discipline:	BIOTECHNOLOGY		
Program/Discipline--Element/Subdiscipline:	BIOTECHNOLOGY--Biomolecular physics & chemistry		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	mark.s.anderson@jpl.nasa.gov	**Fax:**	FY
PI Organization Type:	NASA CENTER	**Phone:**	818 393-5011
Organization Name:	NASA JPL		
PI Address 1:	4800 Oak Grove Drive		
PI Address 2:	M/S: 125-109		
PI Web Page:			
City:	Pasadena	**State:**	CA
Zip Code:	91109-8099	**Congressional District:**	27
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	00-HEDS-02
Start Date:	05/01/2001	**End Date:**	05/01/2004
No. of Post Docs:	0	**No. of PhD Degrees:**	
No. of PhD Candidates:	0	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA HQ
Contact Monitor:	Fung, Paul	**Contact Phone:**	650-604-1276
Contact Email:	pfung@mail.arc.nasa.gov		
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Davidson, Eric (Caltech) Otim, Ochan (Caltech)		
Grant/Contract No.:			
Performance Goal No.:			
Performance Goal Text:			
Task Description:	This task is to apply a newly developed JPL instrument, the Raman-Atomic Force Microscope (Raman-AFM), to biological systems. The Raman-AFM combines the analytical power of Raman spectroscopy with the high spatial resolution of the atomic force microscope (AFM). The basis of this technique was first demonstrated at JPL (M. S. Anderson, Appl. Phys. Lett. 76, 3130, 2000) and has the potential for molecular-resolution imaging spectroscopy. The goal of this work is to use the Raman-AFM to provide chemical analysis of biological macromolecules at very high spatial resolution. The RAFM is a fundamental tool for research. It provides numerous applications for fundamental biology, biological-mineralogical applications and chemical microscopy. The RAFM is a fundamental tool for research. The RAFM provides a combination of nanometer scale imaging and molecular spectroscopy. This has numerous applications for fundamental biology, biological-mineralogical applications and chemical microscopy.		

Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	No progress provided.
Bibliography Type:	Description: (Last Updated: 01/31/2019)
Articles in Peer-reviewed Journals	Anderson MS. "Enhanced infrared absorption with dielectric nanoparticles." Applied Physics Letters. 2003 Oct;83(14):2964. https://doi.org/10.1063/1.1615317 , Oct-2003